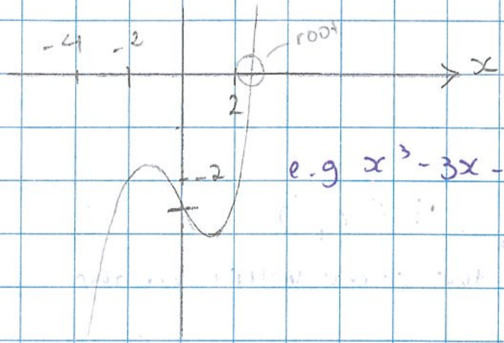
NUMERICAL METHODS

PURE

NUMERICAL METHODS FOR SOLVING EQUATIONS

Some equation cannot be solved using conventional algebra:



e.g $x^3 - 3x - 5$

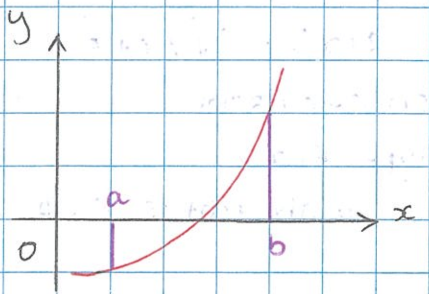
'Numerical methods' of solving means 'trial and error'

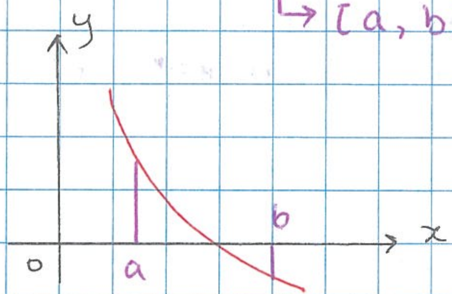
i.e try - check - improve - try - check - improve

THE SIGN- CHANGE RULE

If $f(x)$ is continuous in an interval $a \le x \le b$, and if $f(a)$ and $f(b)$ have opposite signs, then $f(x)$ has at least one root between a and b

→ $[a, b]$ $[2, 5]$
$2 \le x \le 5$





Eg. show that $f(x) = x^3 - 3x - 5$ has a root in the interval $[2, 3]$

Show that $f(2)$ and $f(3)$ have opposite signs

$f(2) = (2)^3 - 3(2) - 5$
$= 8 - 6 - 5 = -3$

$f(3) = 3^3 - 3(3) - 5$
$= 27 - 9 - 5 = +13$

Since $f(2)$ and $f(3)$ have opposite signs, $f(x)$ must have a root with $[2, 3]$

The $x = f(x)$ method

To solve the equation $f(x) = 0$.

1. Rearrange $f(x) = 0$ into $x = F(x)$
2. choose a suitable starting value for x call this x_1
3. Calculate $F(x_1)$ → capital F
4. Let $x_2 = F(x_1)$
5. Calculate $F(x_2)$
6. Let $x_3 = F(x_2)$
7. Calculate $F(x_3)$
8. continue this process letting $x_{r+1} = F(x_r)$
9. Stop when x_{r+1} and x_r have the same value to the required number of decimal places

Example 1: solve $x^3 - 3x - 5 = 0$ correct to 2 d.p. Give the result of each iteration.

$f(x) = x^3 - 3x - 5$

1. $x^3 - 3x - 5 = 0$

$x^3 = 3x + 5$

$x = \sqrt[3]{3x+5} \rightarrow F(x)$

Trick: use $\boxed{\text{ANS}}$ key

$F(x) = \sqrt[3]{3x+5}$

on calculator $\boxed{2}\ \boxed{=}$ (Starting value)

Type $F(x)$

$\boxed{\sqrt[3]{\ }}\ \boxed{(}\ \boxed{3} \times \boxed{\text{ANS}} + \boxed{5}\ \boxed{)} = 2.2239...$

=

=

=

Keep pressing =
It will give the ans

2 $x_1 = 2$

	x	$F(x)$		
3	$x_1 = 2$	$F(2) = 2.223980091$		
4	$x_2 = 2.223980091$	2.2684 5	2.27	2 d.p
6	$x_3 = 2.2684$	2.277 7	2.28	
	$x_4 = 2.277$	2.279	2.28	
9	$x_5 = 2.279$	2.279	2.28	

So, the root is $x = 2.28$ (2dp)

Example 2 :

(i) show that the equation $x^3 - x - 2 = 0$ has a root between 1 and 2

(ii) The equation is rearranged into the form $x = F(x)$ where

$$F(x) = \sqrt[3]{x+2}$$

use the iterative formula suggested by this rearrangement to find the value of the root to 3 decimal places

(i) f(1) and f(2) have opposite signs

$f(1) = (1)^3 - 1 - 2 = -2$ $\quad$ $f(2) = 2^3 - 2 - 2 = 4$

f(1) and f(2) have opposite signs so f(x) has a root in [1, 2]

(ii) $x_1 = 1$ type in calculator

$x_2 = 1.4422$

$x_3 = 1.5099$

$x_4 = 1.5197$

$x_5 = 1.5211$

$x_6 = 1.5213$ $\quad$ So, $x = 1.521$ (3dp)

CONVERGING and DIVERGING VERSIONS OF F(x)

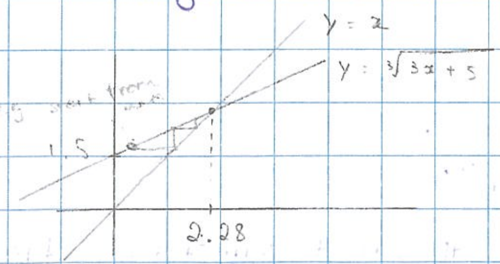
If the values given by the formula F(x) get closer and closer, the formula is said to converge!

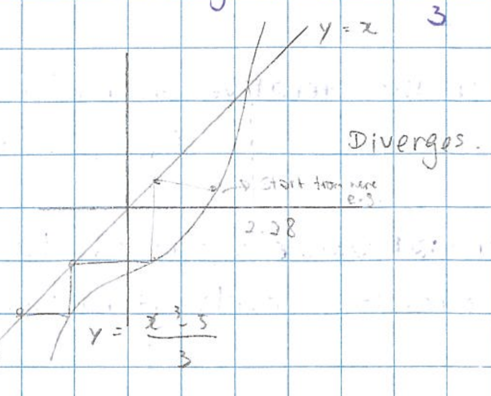
So, the formula $F(x) = \sqrt[3]{3x+5}$ converges!

converge

Solve $x^3 - 3x - 5 = 0$

rearrange to $x = \sqrt[3]{3x+5}$ $\quad$ also rearrange to $x = \frac{x^3 - 5}{3}$





How do you know that formula is converge or diverge? → compare gradien of 2 lines

m < 1 converge $\quad$ m > 1 diverge

$x^3 - 3x - 5 = 0$

it also rearranges to $x = \frac{x^3 - 5}{3}$

F(x) values get further and further apart. This is said to be Diverges.

So, the formula $F(x) = \frac{x^3 - 5}{3}$ diverges

In general, the F(x) formula will converge to the root, if the gradient to F(x) near the root is between 1 and -1 Otherwise, diverges

Generally these formulas will converge:

- square roots, cube roots, 4^{th} roots etc
- logs and natural logs
- e^{-x}
- Inverse trig

using graphs to find an approximate location of the root (a common Cambridge question)

Example:

1. (i) By sketching a suitable pair of graphs, show that the equation $2\cot x = 1 + e^x$, where x is in radians, has only one root in the interval $0 < x < \frac{1}{2}\pi$.

(ii) verify by calculation that this root lies between 0.5 and 1.0

(iii) show that this root also satisfies equation

$$x = \tan^{-1}\left(\frac{2}{1+e^x}\right)$$

(iv) use the iterative formula

$$x_{n+1} = \tan^{-1}\left(\frac{2}{1+e^{x_n}}\right),$$

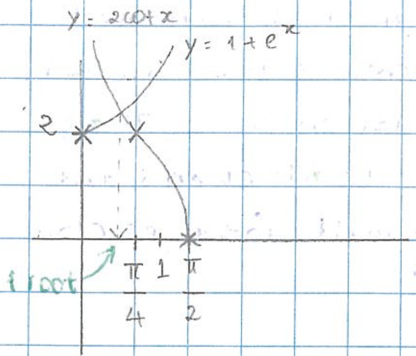
with initial value $x_1 = 0.7$, to determine this root correct to 2 d.p. Give the result of each iteration to 4 d.p.

Ans

(i) $f(x) = 0$

$2\cot x - 1 - e^{x} = 0$

$y = 2\cot x$
$y = 1 + e^{x}$



(ii) $2\cot x - 1 - e^{x} = 0$

$f(0.5) = \frac{2}{\tan(0.5)} - 1 - e^{0.5} = 1.0122541.$

$f(1) = \frac{2}{\tan(1)} - 1 - e^{1} = -2.434096..$

Since $f(0.5)$ and $f(1)$ have opposite signs, So, there must be a root in $[0.5, 1]$

(iii) $x = \tan^{-1}\left(\frac{2}{1+e^{x}}\right)$

$\frac{2}{\tan x} = \frac{1+e^{x}}{1}$

$\frac{\tan x}{2} = \frac{1}{1+e^{x}}$

$\tan x = \frac{2}{1+e^{x}}$

$\cancel{\tan}\, x = \tan^{-1}\left(\frac{2}{1+e^{x}}\right)$

(iv) $\boxed{0.7}$ $\boxed{=}$

$\tan^{-1}(2 \div (1 + e^{\wedge} ANS)) =$

$x_1 = 0.7$
$x_2 = 0.5859$
$x_3 = 0.6208$
$x_4 = 0.6101$
$x_5 = 0.6134$
$x_6 = 0.6124$
$x_7 = 0.6127$
$x_8 = 0.6126$

So, $x = 0.61$ (2dp)